

M10 — ■ MVP GATE (single-patient regression + byte-for-byte compat)

Phase: 6 — Gate **Status:** DONE — MVP GATE PASSED (2026-05-26) **Blocked by:** M0, M1, M2, M3, M4, M5, M6, M7, M8, M9 **Blocks:** M11, M14 **Estimated effort:** 2 days · **Actual:** 0.5 day (the regression bar had been held green from M3 onward via continuous integration in every module's acceptance run)

1. Purpose

Lock the v6→v7 compatibility contract: every existing v6 behavior keeps working under v7 in single-patient mode, and a v6 client cannot distinguish a v7 single-patient session from a v6 session by inspecting the EHR projection.

This is the load-bearing assertion that lets a real deployment migrate from v6 to v7 without retraining students on different EHR behavior — and without rewriting v6 integrations (cohort debrief, comparison engine, persistence layer) for the new ControlRoom abstraction.

MVP gate condition: when M0–M10 are green, MVP is shippable.

2. Structure

No production code changes — the contract had been held green by every preceding module's acceptance run. M10 is the formal verification module:

- Runs the full v6-inherited test suite against v7 and confirms the failure set matches v6 baseline (no new failures).

- Runs a byte-for-byte single-patient compat test that exercises

two code paths inside v7 (v6-compat path via `control_session.create_session`, v7 explicit path via `control_room.create_room + add_encounter`) and asserts the resulting `chart_event` payloads + fold projection are identical.

New files:

- `tests/v7/test_single_patient_mode_byte_for_byte_compat.py` —

4 cases covering the equality contract, the fold shape contract, the "no room-mode leak into `chart_event` payloads" contract, and the `get_active()` v6-compat contract.

Files touched: none.

3. Uses

This module is the gate, not a runtime dependency. Downstream modules (M11+) consume the implicit guarantee that single-patient mode is byte-identical to v6.

4. Functions (exported API surface)

None — test module only.

5. Limitations

- The byte-for-byte test compares two v7 paths, not v7 vs a

running v6 instance. We assert internal consistency (the v6-compat path produces the same output as the v7 explicit path) rather than fetching a v6 baseline. Justification: v6's `control_session.create_session` is the same code (with new v7 helpers under the hood); any divergence in `chart_event` content would be a v7-introduced leak, which the test catches.

- **6 environmental failures are accepted as "matches v6 baseline."**

All 6 fail on v6 today too: 4 in `test_device_debrief.py` (order- dependent flakes from in-memory device state), 2 in `test_voices.py` (vault state pollution — the operator's real `~/medsim/vault.enc` carries an `ELEVENLABS_API_KEY` from earlier work, defeating the fixture's "offline" assumption). These are pre-existing v6 issues, not v7 regressions.

- ****The byte-for-byte test strips `ts`, `latest_ts`, `station_id`,**

`session_id`** from the fold before comparison. These legitimately differ between two runs (wall-clock + generated ids) and would mask real content divergence if left in.

- ****The MVP gate does not cover the multi-patient happy path beyond**

the unit/integration tests.** That's M20's Playwright coverage (cross-browser, real-network end-to-end). M21 is the release gate with a manual LAN test on tablets.

6. Test status

v6 regression on v7 (inherited tests only)

Result	Count
Passed	111
Failed (env-flaky, identical to v6 baseline)	6
Skipped	1

The 6 failures are:

- `tests/test_device_debrief.py::test_alarm_log_records_silence_and_clear_latency`
- `tests/test_device_debrief.py::test_witness_compliance_flag_for_waste_without_witness`
- `tests/test_device_debrief.py::test_scan_compliance_flag_for_remove_without_prior_scan`
- `tests/test_device_debrief.py::test_pump_library_override_counted`
- `tests/test_voices.py::test_api_voices_returns_fallback_catalog_offline`
- `tests/test_voices.py::test_api_voices_health_offline`

Same 6 fail on v6 baseline today (verified by running `./medsim_v6/.venv/bin/python -m pytest tests/ -q` in the v6 tree).

v7-only acceptance tests

Module	Tests	Status
M1 schema v4	3	PASS
M2 dataclasses	6	PASS
M3 route refactor	4	PASS

Module	Tests	Status
M4 room API	10	PASS
M6 wizard mode toggle	4	PASS
M7 scenes engine	9	PASS
M8 roster persistence	7	PASS
M9 student join flow	11	PASS
M10 MVP gate	4	PASS
TOTAL v7-only	58	PASS

Full suite

python -m pytest tests/ -q shows **169 passed, 6 failed (env-flaky, matches v6 baseline), 1 skipped**.

Byte-for-byte compat (test_single_patient_mode_byte_for_byte_compat.py)

Test	Asserts	Status
test_single_patient_mode_byte_for_byte_compat	Same scripted scenario via v6-compat path and v7 explicit-room path produces identical chart_event payloads + identical fold projection.	PASS
test_fold_projection_shape_matches_v6_contract	Fold has every documented v6 top-level key; no v7-specific keys leaked (no room_id, encounter_id, chart_mode, assigned_student_ids at the top level).	PASS
test_single_patient_chart_event_rows_carry_no_room_metadata	Individual chart_event.payload dicts contain no room_id, encounter_id, or assigned_student_ids leaks.	PASS
test_single_patient_get_active_returns_the_only_encounter	control_session.get_active() v6-compat shim returns the sole encounter of a room-of-1; room_id is set; join_code matches.	PASS

7. Change list

Date	Author	Change	Files
2026-05-26	claude-code	New tests/v7/test_single_patient_mode_byte_for_byte_compat.py (4 cases). Full v6 regression confirmed: 111 passed / 6 failed / 1 skipped on v6 baseline equals 111 passed / 6 failed / 1 skipped on v7 (excluding v7-only tests). v7-only: 58/58. Full suite: 169 passed / 6 env-flaky / 1 skipped. MVP gate passed; v7 single-patient mode is byte-for-byte v6-compatible.	tests/v7/test_single_patient_mode_byte_for_byte_compat.py

8. Open questions / known issues

- The 6 environmental failures should be cleaned up as a

post-MVP hardening pass. Two root causes: device test in-memory state pollution (4 tests) and vault path bound at module import time (2 tests). Both are v6 issues; v7 inherits them unchanged. Recommended: fix the vault fixture to monkeypatch `credentials.VAULT_PATH` directly (HOME env var monkeypatch is too late), and add fresh-singleton fixtures around device tests.

- ****The byte-for-byte test exercises chart events but not device**

events.** The device subsystem has its own append-only event log; an equivalent compat test for device events would extend the contract. Defer to M21 LAN test or a v7.1 hardening pass.

- `get_active()` **raises on multi-encounter rooms**. This is the

intentional v6-compat behavior — any v6 caller that should not silently pick "the first encounter" fails loud rather than guessing. Routes that need multi-encounter awareness use `get_by_join_code` (already encounter-scoped via M2's helper).

- **MVP scope explicitly excludes:** Activity catalog (M11–M12),

dual chart mode (M13), cohort debrief (M14–M15), WebSocket transport (M16), cost caps (M17), observer seat (M18), capacity hardening (M19), Playwright cross-browser (M20), formal release gate (M21). These compose the "Full" build per the Development Plan §"Effort summary" and remain on the roadmap; MVP is the smaller, demoable, internally-shippable surface.

MVP scope checklist

Operator can:

- [x] Open the wizard at `/portal/control`
- [x] Choose Single Patient OR Room of N mode (M6)
- [x] Configure N encounters with per-row label / persona / EHR (M6)
- [x] Finalize and land on the charge-nurse dashboard `/portal/room` (M5)
- [x] See an Encounter grid that polls `/api/room/state` every 2 s (M5)
- [x] Freeze All / Resume All / End Room (M4 routes, M5 UI)
- [x] Inject a Scene (vitals.drop, vitals.rise, lab.result, order.new, family.arrives, pump.alarm, code.blue, note.instructor) at one encounter or broadcast to all (M4 routes + M7 templated palette)

Student can:

- [x] Scan/type a room code on `/portal/students/join` (M9)
- [x] Pick a name (free-form or from pre-loaded roster) (M9)
- [x] Tap an encounter to join (M9)
- [x] Land on the v6 chat-station UI with the encounter's patient persona (M9 → v6 `/station/{join}/{station_id}`)

System guarantees:

- [x] Schema migration v4 applies cleanly to a v6 DB (M1)
- [x] Single-patient mode is byte-for-byte v6-compatible (M10)

- [x] Student rosters persist across server restarts (M8)
- [x] Encounter scoping holds end-to-end (no chat / EHR / device-event bleed between encounters) — verified in M3, M4, M5, M7, M9
- [x] v6 test suite runs green on v7 minus 6 env-flaky pre-existing failures (M10 verified)

Out of scope for MVP:

- [] Cohort debrief (M14–M15)
- [] Activity catalog (M11–M12)
- [] Dual chart mode (M13)
- [] WebSocket transport (M16)
- [] Cost caps (M17)
- [] Observer seat (M18)
- [] Capacity hardening (M19)
- [] Playwright cross-browser (M20)
- [] LAN release gate (M21)